Hewlett - Packard, Inc.
Long Term Debt

Hewlett-Packard, Inc. (HP) is California-based multinational company focused on developing and manufacturing computer hardware, software and technology consulting services. The company was founded in a garage in Palo Alto, California, by Bill Hewlett and Dave Packard in 1939 and had an IPO in November, 1957 at $16.00 per share; as of December 2017, HP's market capitalization was about $35.51 billion.

HP heavily relies on debt financing. Its long-term debt (the sum of its current portion and long-term portion of its long-term debt) at the end of 2016 was $6.8 billion.

Answer the following questions using the excerpts from Hewlett-Packard's annual report for the year ended October 31, 2017:

1. What is the amount of HP's long-term debt at the end of fiscal year 2017?

   Answer:

   Current portion of long-term debt: $96 (Note 11);

   Long-term debt: $6,747 (Note 11 or B/S)

   Total long-term debt 6747 + 96 = $**6,843**

2. What is the face value of HP's long-term debt (including capital lease obligations)? On average, did HP issue its long-term debt at par, discount or premium?

   Answer:

   Face value = $7,836

   On average HP issued its long-term debt at a discount

3. What is HP's interest expense for 2017?

   Answer:

   $309 (Note 7)

4. How much cash did HP pay for interest in 2017?

   Answer:

   $322 (Supplemental cash flow disclosures under CF Statement)

5. What is the non-cash component on HP's interest expense in 2017? Is it amortization or discount or amortization of premium? Suppose that cash paid for interest in 2017 includes 22 million related to interest due in 2016.

Answer:
$309-($322 - $22) = $9 (Amortization of discount)

6. For each of the following bond issues, determine the face value, the coupon rate, the net book value as of October 2017, and whether the bond was issued at a discount, premium, or at par. Assume that for these notes no additional debt was retired or issued during fiscal year 2017.

	Face Value	**Coupon Rate**	**Net BV Oct 2017**	**Discount/ Premium/ Par**	**Historical Discount Rate**
US Dollar Global Notes (due December 2020)	1,350	3.75%	648	Discount	>≈3.75%
US Dollar Global Notes (due June 2021)	1,250	4.3%	1,249	Discount	> 4.3%
$750 US Dollar Global Notes (due January 2019)	750	USD LIBOR plus 0.94%	102	Par	USD LIBOR plus 0.94%

7. Answer the following questions pertaining to the US Dollar Global Notes due June 2021:

a. What is the interest expense HP recognized pertaining to this bond?

Answer:
Coupon payment = 1,250 × 4.3% = 53.75
Amortization of Bond Discount = ΔNBV = 1,249 – 1,248 = 1
Interest expense = 53.75+1=**54.75**

b. What is the historical interest rate for the bond?

Answer:
Interest expense = BB of NBV * Discount rate
Discount rate = $54.75/$1,248 = 4.387%

8. Suppose that HP had a zero-coupon bond and made the following disclosure about it:

	2017	**2016**
$500 m zero-coupon bond due in 2028	356	345

Answer the following questions pertaining to US Dollar Zero Coupon bond due in 2028:

a. What is the interest expense HP recognized pertaining to this bond?

Answer:
Coupon payment = 0
Amortization of Bond Discount = ΔNBV = 356-345 = 11
Interest expense = 11

b. What is the historical interest rate for the bond?

Answer:
Interest expense = BB of NBV * Discount rate
Discount rate = $11/$345 = 3.19%

9. Footnote 9 of HP's 2017 10K contains the following information:

"HP estimates the fair value of its debt primarily using an expected present value technique, which is based on observable market inputs using interest rates currently available to companies of similar credit standing for similar terms and remaining maturities, and considering its own credit risk. […] The estimated fair value of HP's short- and long-term debt was $8.1 billion at October 31, 2017 compared to its carrying amount of $7.8 billion at that date."

Using the above information, answer the following questions:

a. Is the estimated value of debt greater than, smaller than or the same as the NBV? What explains the difference? How does GAAP reflect this?

Answer:
Estimated fair value of Debt = $8.1 billion
NBV of Debt = $7.8 billion
NBV < Fair value ⇒ Interest rates ↓
GAAP ignores fluctuations in fair market value under effective interest rate method.

b. Suppose HP repurchased all of its debt. What would be the effect on HP's net income?

Answer:
Loss on retirement of debt = fair market value – NBV = $0.3 billion
Net Income would decrease by $0.3 billion.